

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000805		
	Version:	5.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	19-Dec-2014
Title:	SM DG57-DG517. Robot module verification			

DG-57 / DG-517 Service Manual

ROBOT MODULE VERIFICATION

Procedure description
This procedure describes how to check the Robot module parts: home detectors, positioning performance, card transport, level detection and impact detection. Note: Document REGD-0000805_v5 replaces document REGD-0000805_v4.

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EQUIPMENT

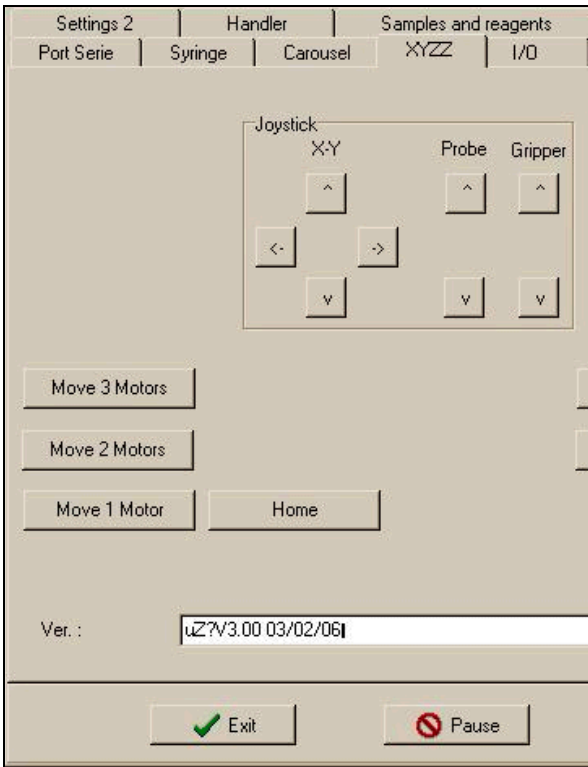
1. An instrument connected with a PC with the software installed

SET UP

1. Turn on the instrument.
2. Execute the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).

PROCEDURE 1 - Checking homes

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'XYZZ' tab in ST software.  Picture 1	The software looks like Picture 1.

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Process	Expected Results
2. Press 'Home' button. (see Picture 1) Picture 2	The Picture 2 window appears.
3. Select 'Probe' and press 'Ok' button. (See Picture 2)	The probe moves slowly up until home is detected, then down until home is not detected and finally up to home again.
4. Select 'Gripper' and press 'Ok' button. (See Picture 2)	The gripper moves slowly up until home is detected, then down until home is not detected and finally up to home again.
5. Select 'Y' and press 'Ok' button. (See Picture 2)	The robot head moves slowly forward until home is detected, then backward until home is not detected and finally forward to home again.
6. Select 'X' and press 'Ok' button. (See Picture 2)	The robot head moves slowly right until home is detected then left until home is not detected and finally right to home again.
7. If a 'Home' movement fails, check the 'Home' sensor and cables.	---

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PROCEDURE 2 - Positioning performance

Process		Expected Results
1. Access 'Handler' tab in ST software. Picture 3		The software looks like Picture 3.
2. Place a card in incubator first position.		---
3. Press 'Remove card from incubator' button. (see Picture 3) Picture 4		A window appears. (See Picture 4).
4. Type 1 and Press 'Ok' button. (See Picture 4)		The arm takes the card.
5. If the gripper can't pick up the card, check gripper adjustment (see document REGD-0000807).		---
6. Press 'Place card into incubator' button. (see Picture 3)		A window appears. (See Picture 4).
7. Type 1 and Press 'Ok' button. (See Picture 4)		The arm places the card.
8. Press 'Random movements' button. (see Picture 3)		The instrument arm starts to move randomly.

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Process	Expected Results
9. After 6 minutes around 500 movements have been done. Press 'Stop' button. Picture 5	A window appears showing the number of movements. (See Picture 5).
10. If any error appears, check motor and encoder cables.	---
11. Place a card in incubator first position.	---
12. Press 'Remove card from incubator' button. (see Picture 3)	A window appears. (See Picture 4).
13. Press 'Ok' button. (See Picture 4)	The arm takes the card.
14. If the gripper can't pick up the card, check motor and encoder cables, transmission belts and pulleys.	---

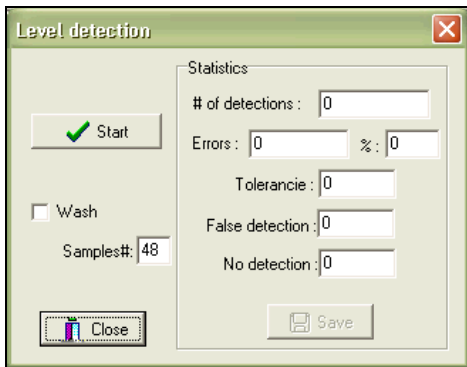
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PROCEDURE 3 - Gripper performance

Process		Expected Results
1. Access 'Procedures' tab in ST software. Picture 6		The software looks like Picture 6.
2. Fill the incubator with 24 cards.		---
3. Press 'Loading centrifuge' button.		The arm begins to load the centrifuge with cards.
4. The instrument cycles loading, centrifuging and unloading the centrifuge until the 'Stop' button is pressed.		---
5. Let the test run for 1 hour		---
6. Press 'Stop Button'.		The instrument stops when the cycle finish.
7. If the test fails before the 1 hour cycle check the gripper calibration. See documents REGD-0000807 and REGD-0000808.		---

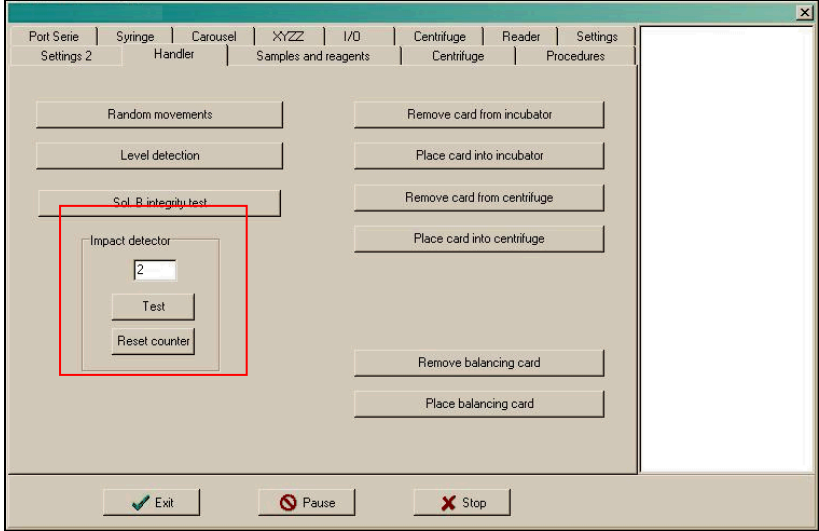
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PROCEDURE 4 - Probe performance

Process	Expected Results
1 Access 'Handler' tab in ST software.	The software looks like Picture 3.
2 Ensure that wash solution bottles have at least 0.5 liters and waste bottle are not full.	---
3 Ensure that all bottles racors are properly connected.	---
4 Press 'Level detection' button	The Picture 7 window appears, the pumps turn on, the arm places on the dilution cup and short priming is done.
 Picture 7	
5 Place 3 15 mm tubes, with 3 different levels of saline 0.9% water in the 3 first positions of the samples carrousel.	---
6 Type 3 in the 'Samples#' box.	---
7 Press 'Start' button. (See Picture 7)	A level testing cycle starts, detecting in the 3 tubes.
8 Wait until 500 detections are done. Press 'Stop' button.	---
9 Check no errors where detected during the process.	---

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PROCEDURE 5 – Impact detection.

Process		Expected Results
1 Access 'Handler' tab in ST software. 		The software looks like Picture 8.
2 Press 'Reset counter'.		Counter will be set to 0.
3 Press 'Test'.		Probe will go down to impact against incubator surface. Impact detector counter will increase to 1.
4 Repeat step 3 twice.		Probe will go down again to impact against incubator surface. Impact detector counter will increase until 3.
5 If impact detection counter doesn't arrive until '3' it means that impact detection assembly doesn't work correctly.		Check impact detection assembly.

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	Document's reference changed updated ST software windows.	Document REGD-0000805_v2 replaces document 57033V0M000.
3.0	2010/09/10	Updated all the pictures.	Due to the changes introduced in last versions, new functionalities have been added to the technical service software.
4.0	June 2011	Updated the Document Change History: -Date format changed to YYYY/MM/DD -Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	June 2011	Document REGD-0000805_v4 replaces document REGD-0000805_v3.	Document version changed.
4.0	June 2011	Added procedure 5 'impact detection'.	Added impact detection checking.
4.0	June 2011	Document title.	Commercial reference substituted by project reference.
4.0	June 2011	Document header.	Increase information to easy document search.
5.0	December 2014	Gripper performance	Length of the test is modified

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